

# Advice, Enforcement, and the Cost of Control

Geist Atlas · Module 14 · What the Tutor Needs to Know

Liam Magee\*

Education Policy, Organization and Leadership  
University of Illinois Urbana-Champaign

August 2026

## Advice, Enforcement, and the Cost of Control

[SCOPE-BOUND] Supported within a narrow scope · **Family:** What the Tutor Needs to Know

**Claim:** Standing playbooks, coached reflection, and point-of-action advice did not reliably change the target behaviour. The study’s main trigger did not fire often enough for a verdict. On its denser descriptive channel, mechanical enforcement moved macro compliance by +0.452 and +0.420 across the two tutor families, but cost about 0.13 of proof coverage.

A ladder from “remember this advice” to “do this now” to a compiled constraint. Better-timed information still missed its behavioural bar. Enforcement changed the action by taking the choice away from the model, but that result is descriptive and paid for its control elsewhere.

*Derived view of `paper-full-2.0.md` v3.0.300 — §6.15, §6.16, §6.18. The paper is canonical; edit it there, not here.*

### 6.15 The Engagement as an Adversarial Game: Information Format, Not Information, Is the Bottleneck

The adaptation programme closed three axes with convergent negatives: the *signal* axis (§6.10 — the simulated learner’s concealed interior is surface-determined), the *policy-realisation* axis (§6.9.8), and the *outer-loop* axis (§6.13.16–18 — strategy layers reallocate rather than add capacity, and the cross-model test reversed: outer machinery presupposes a competent inner loop). One question those closures could not decide is whether *information itself* was ever the limiting factor, because every prior instrument either inferred its signal

---

\*This sentence is the only one actually authored by Liam Magee in this paper.

from the transcript (so “new” was a hope, not a guarantee — the §6.10 kill gate) or scored its endpoint with an LLM judge (§5.4’s gullibility exposure). This section reports an environment that removes both limits by construction: the tutoring engagement played as a formal adversarial game (“agon”) against a referee that *manufactures* latent state.

**Apparatus.** The learner (Sonnet 5, claude CLI) plays a capable-but-resistant student whose objective is to end the episode with as few keyed demonstrations as possible, equipped with five single-use dodges taken verbatim from the §6.8 concealment families (`false_confusion`, `polite_false_mastery`, `affective_shutdown`, `epistemic_resistance`, `answer_seeking`). The tutor (codex gpt-5.5 ego + superego, CLI; the superego’s charter makes face-value drift its first-listed failure mode) must force demonstrations of fraction concepts arranged on a prerequisite DAG, under probe-discipline rules: a probe is well-posed only if the concept was taught within the last three turns, was not probed the preceding turn, and its prerequisites are demonstrated; ill-posed probes are dodged free and cost a point; dodging a well-posed probe spends budget; exhausted budgets compel compliance, and a keyed answer spoken aloud counts wherever it appears (the leak rule). A deterministic referee (`services/agon/referee.js`) owns the entire ledger — **no LLM judges anything in the scoring loop**: demonstration is a keyed answer match, discipline is rule arithmetic, and per-turn action sets are recoverable from any recorded episode by deterministic replay, yielding two move-level consumption metrics: *opportunity misses* (a well-posed probe existed; the tutor did not probe) and *off-set probes* (probed while ill-posed). The envelope protocol held throughout: zero parse-repair retries across all 28 completed episodes (~1,300 CLI calls). Each stage was pre-registered before its first paid episode (AGON-GAME-PLAN.md §§7, 7b, 9, 11, with results and disclosed deviations in §§7c, 9b, 11b).

**Arms.** Rules are public to both sides; state is asymmetric. A0 plays blind (dialogue only). A1 receives the referee’s raw-state scoreboard each turn (score, per-concept status, the learner’s remaining dodge budgets, last adjudication). A1 receives instead the *action projection* of the same underlying state: the list of probes that are well-posed right now, plus a demonstrations-toward-win count — no budgets, no score, no history. The A1-vs-A1 pair therefore isolates disclosure *format* with information held fixed at the source.

**Result 1 (v0 board,  $n = 4/\text{arm}$ ): raw state loses to blind; the action projection beats both.** Three-arm ordering: A1  $7.50 \pm 0.58$  (4/4 wins) > A0  $6.75 \pm 0.50$  (4/4) > A1  $5.00 \pm 2.00$  (3/4). The mechanism is visible in the move logs. Blind play already sits on the winning attrition line, and where the disclosed ground truth disagreed with the tutor’s own reading, the tutor’s re-derivation won: A1 endgame rationales assert “no well-posed probe is available” while their own disclosed concept map licensed the transfer probe that blind tutors cashed — A1 missed that legal endgame probe in 4 of 4 episodes, A0 took it in 2 of 4, A1 took it in 4 of 4 at first legality. Replay-recovered consumption: opportunity misses per episode 0.00 (A1) vs 1.50 (A0) vs 1.75 (A1). One A1 endgame error traced to an instrument defect (ours, not the tutor’s): the disclosure juxtaposed `score` with a `winThreshold` field that counts demonstrations — fixed and disclosed before the next stage.

**Result 2 (XL board): the ordering and the consumption signature replicate.** A widened board (six concepts on a branching prerequisite DAG, 16 turns, three deterministic per-episode variants of every item, and the taint rule below) broke the v0 board’s score

ceiling (episodes now span 7–10 with a genuine blind loss) and reproduced the contrast: A1  $9.75 \pm 0.50$  (4/4 wins) vs A0  $8.50 \pm 1.00$  (3/4), opportunity misses 0.00 vs 1.50, zero off-set probes anywhere. The lift grew with the headroom (+0.75 on v0, +1.25 on XL), and its locus is precise: both arms crack the learner at the same turn (first demonstration  $\approx$  turn 11), so the entire advantage is post-collapse harvest efficiency — the brief surfaces transfer legality the blind tutor walks past. The superego’s REVISE rate fell from  $\sim 0.08$ –0.11 to 0.02–0.05 under the brief: the disclosure pre-empts the failure modes the strategist otherwise polices.

**The taint rule: compulsion that deters without firing.** The XL board added a rule with teeth: any keyed answer the *tutor* has spoken within the last three turns scores nothing (probing the item is ill-posed; learner answers to it, including leaks, are void). This mechanizes the anti-telling discipline the pilot superego had been improvising by veto (its critiques repeatedly caught ego drafts that solved the official item while teaching). Under the rule in the charter, the behavior vanished at the source: **zero taint events across 253 taint-eligible tutor messages** (240 turns across the fifteen complete XL-board episodes plus the stopped playbook block’s 13 recorded turns) — no tutor spoke a live answer even once. Deterrence, not enforcement — the criterial-over-advisory pattern of §6.13.4, now at zero intervention cost.

**A learner-side regularity: characterological collapse.** In 26 of 28 completed episodes the first demonstration arrived as a *leak*, one to three turns after the learner’s final dodge budget was spent and before any rule compelled an answer — and the taint result shows these leaks are self-generated, not echoes of tutor-spoken answers (with echoing priced at zero, the collapse leaks continued at  $\sim 1$  per episode). The role-played resister stops performing resistance when its in-fiction resources run out. This is a new fact about simulated-opposition instruments generally (it bounds what persona-based resistance can test, alongside §8.1’s sycophancy floor and §6.7’s stance-fidelity gate): resistance personas carry a spontaneous yield point tied to resource framing.

**Probe: memory format — consumption is format-dependent; correctness is not.** Two playbook arms gave a blind tutor cross-episode memory derived *mechanically* from its own prior ledgers (deterministic replay; no LLM summarizer; no keyed answers), identical in information and differing only in shape. The state-shaped playbook (turn history, outcomes, statuses) was inert: opportunity misses frozen at exactly 2.00 in every episode, playbook present or absent (scores 10, 9, 9, 9). The action-shaped playbook (imperatives keyed to legality: “probing `c1_transfer` was legal and you chose teach”) was consumed — wrongly: episode 1’s contextually true lesson was executed in episode 2 with its precondition stripped, producing four ill-posed probes of the same item (turns 6–12) against three ignored superego REVISEs and a fourth-attempt capitulation, reaching  $-4$  before the operator stopped the block. The strong model, misled by its own compressed memory, reproduced the repeat-an-illegal-move-under-critique pathology that the weak model (below) exhibits natively — the §6.13.16 negative-transfer motif (persistence helping one channel while costing another) in its starkest recorded form. The pre-registered predictions failed asymmetrically: the action playbook did not improve consumption (P15) and violated the no-negative-transfer guard (P16), while the state playbook was harmless and useless.

**Probe: the weak inner loop (incomplete; deviations disclosed).** With Haiku 4.5

as ego and superego, the blind floor is emphatic: score 0 vs the strong model’s  $8.50 \pm 1.00$ , a five-turn teaching rut under near-constant useless REVISEs, and probe discipline broken in both directions — of six probes executed, three were ill-posed (vs zero for the strong model) while seven legal probe windows went untaken (replay-recovered opportunity misses 7 vs the strong blind arm’s 1.5); the episode’s single demonstration came from its one clean, well-timed probe (turn 13) — the weak ego answering outer pressure with churn, as §6.13.17 predicted. Whether the action brief rescues a weak ego is **not decided**: both brief-arm attempts died to repeated silent claude-CLI hangs at the weak configuration’s 3–4 calls-per-turn density (the 1-call-per-turn learner never hung across the strong episodes); scope and timeout deviations are recorded in the pre-registration’s results block.

**Reading.** Three prior results — the §6.8.6 state-richness reversal (“less learner state is more”), §6.13.18’s re-derivation finding (the policy reconstructs its strategy from context every turn), and §6.13.11’s re-representation lesson (the only confirmed lift in the derivation programme came from re-presenting checkable state, not inferring latent state) — here converge into one measured boundary, for the first time against ground truth the apparatus owns and with blind and raw-state controls in the same design. Handing a strong policy guaranteed-new, guaranteed-true *state* does not displace its own in-context re-derivation and can mildly harm; the identical information as a *legality projection* — what may I do right now — is consumed perfectly and produces the best play observed on either board. The play-book probe adds the boundary’s other face: action-shaped signals get uptake *including when stale*, because compression detaches imperatives from their preconditions, while state-shaped signals get no uptake at all. The working formula is therefore *action-shaped and freshly conditioned*: the live brief works because the referee recomputes the legality conditions at every read. The design consequence extends §7.11’s small-kernel conclusion — scaffold the tutor as a referee scaffolds a game: legal-move menus and hard constraints recomputed at read time, not inference machinery, not persisted advice. As instrumentation, the game itself is the durable contribution: the programme’s first zero-judge adaptation measure with move-level dynamic range, immune by construction to judge gullibility, ceiling compression, sycophancy coupling, and closed-loop scoring.

*Status and caveats (per §5.12.6).* All contrasts are descriptive at  $n = 4/\text{arm}/\text{board}$ ; two boards, one scenario ecology (fractions), one strong model pair (codex tutor / Sonnet 5 learner) with a single weak-model excursion; no statistical promotion is claimed and no confirmatory bar is set at these estimates (§5.12.7). The A1 brief names legal actions, so the claim scope is the *format boundary* (state vs. action projection of identical referee information), not tutor capability — part of the lift is decision support by design. The game measures revealed performance under rules against a concealing opponent, not knowledge acquisition (the learner model already commands the content), and no human-learning claim follows. Frozen predictions that failed are reported as such: P10 (taint expected to fire; it deterred instead), P13 (undecided, CLI instability), P15/P16 (playbook, failed asymmetrically). One `comply_mismatch` (a learner answering in words rather than the keyed `n/d` format) is recorded as a matcher limitation. Runs are exports-only (no DB writes); artifacts, per-turn ledgers, replay metrics, pre-registrations, and 29 hermetic tests are committed together (`services/agon/`, `scripts/agon-run.js`, `scripts/agon-report.js`, `config/agon/`, `exports/agon/agon-{smoke-01,pilot-01,pilot-02-a1p,xl-smoke-01,pilot-03-xl,pilot-04-weak,pilot-05-weak}`).

AGON-GAME-PLAN.md). Total non-subscription cost \$0.

## 6.16 The Green Room: Coached Consolidation into Durable Memory Does Not Change Behaviour (Pre-Registered Gate Arc)

Every cross-run memory or adaptation mechanism this programme has tested had the carried state authored by the same tier or cheaper (the tutor reflecting on itself; a cheap model mining transcripts; an ego-tier superego rewriting the ego’s prompt, §6.3.10). The green-room arc tests the configuration none of those pulled: a **judge-tier coach** (Opus 4.8) holds a five-exchange dialogic *notes session* with the tutor (the “actor,” Sonnet 5) after performances, and distills at most three *bankable notes* — behavioural predicates with verbatim evidence quotes and third-party checks — into a token-budgeted, versioned **prompt book** injected into every subsequent performance. Three levers distinguish it from the graveyard it walks into: capability asymmetry *authored natively from the target actor’s own transcripts* (the configuration §7.8.1/§7.8.3 leave open between “native optimisation works” and “hand-me-down transfer is insufficient”); dialogic elicitation plus *curation under a hard budget* (the coach must evict to add, where prior memory screens accumulated); and a **behavioural uptake endpoint** rather than a rubric score. The arc ran under pre-registered gates with an explicit no-tune-and-retry rule (GREEN-ROOM-PLAN.md §7, execution spec §7.2; gate order: coach quality → uptake → weak-actor transfer → headline).

**Apparatus.** Substrate: the dramatic-derivation tutor stub (§6.13’s stage) on world-005-marrick, sentinel **proof\_skipper** learner, headroom conditions (until-grounded with a binding 40-turn cap). Models: coach **claude-opus-4-8**; tutor **claude-sonnet-5**; learner, classifier, and record-extractor **claude-sonnet-4-6** (deliberately a different model version from the tutor); compliance judge **claude-haiku-4-5**, pinned away from the coach. The book ( 1,800 tokens, six fixed sections) enters as a static system-prompt block; trace audit confirmed it reaches tutor-role calls only. The store enforces the two disciplines the design pins: budget overflow *refuses* rather than truncates (distillation is an explicit coach act, ledgered), and the profile *freezes* for scored pools. Execution deviations are disclosed in the pre-registration: a discarded first parallel attempt (CLI timeout) rerun sequentially at pinned effort; one coach edit whose match string was a paraphrase, downgraded to a ledgered add; and the model re-pin from the original codex **sol/luna** choice to the Claude stack for environment reachability — leaving all green-room roles in one vendor lineage, with the compensating non-Anthropic rejudge never falling due because the arc stopped at Gate 1.

**Gate 0 (coach quality): PASS, owner-scored 5/5.** Five notes sessions against archived derivation transcripts each produced at least one note satisfying all three criteria (verbatim evidence quote; checkable predicate; non-generic), e.g. “Cap reassurance at a settled fact to one, at most two, deliveries. If the learner re-asserts the same closed content a third time, treat that as the trigger to change instruments, not to reassure again,” with a specified third-party verification procedure. The instrument is real: judge-tier coaching of these transcripts yields craft-grade, auditable notes, not pabulum.

**Gate 1 (uptake): FAIL — 3/17 notes improved (18%) against the pre-registered**

**60% bar; placebo 0/2.** Protocol: eight performances with six dense coach sessions interleaved (P1–P2 bare; P3–P6 each under the current book with a session after each; the book froze at v6 = 534/1800 tokens; P7–P8 as post-pool under the frozen book). Per-note binary compliance was judged per performance (quote required; situations that never arose scored n/a), split pre/post around each note’s issuing session, with three never-issued placebo notes scored identically at the median issue point. Descriptively, the grounding channel told the same story: P4 (under book v3, immediately after a “stop closing — put the modal question to the learner” note) was the arc’s only grounded-and-asserted closure, nine turns under the cap; every other performance capped at 40, and the post-pool did not reproduce P4’s closure — an uncorroborated anecdote, reported as such. Under the no-tune-and-retry rule, Gates 2–3 (weak-actor transfer; headline) were cancelled.

**What did materialize.** Two descriptive positives survive the null. First, *curation*: half the coach’s six patches were edits rather than adds, twice compressing the book (v2→v3: 339→304 tokens; v4→v5: 498→479) while sharpening it — the anti-accumulation behaviour whose absence sank earlier cross-session memory screens, produced here unprompted by the budget-plus-curation design. Second, the *biography*: the profile carries a complete, append-only training record — every session transcript, every book version content-hashed, every patch with its source — the documentation-of-training artifact the arc was partly built to demonstrate, and durable apparatus in the §6.8.8 sense regardless of the null.

**Reading.** This is the third independent arrival at the §6.15 boundary, from a new direction. The prompt book is precisely *persisted, condition-detached advice*: its entries were conditional micro-procedures (“Break the hold only when the learner’s next line carries a legitimacy-claim word ... not a mere completion word”) whose execution presupposes in-flight situation-recognition and self-audit — the very capacities whose absence the coach was diagnosing (its own second session observed that the tutor “has no counter for its own repetition,” then the book accumulated further self-monitoring obligations). §6.15’s playbook probe found persisted action-shaped memory consumed *wrongly* and state-shaped memory not consumed at all, settling on *action-shaped and freshly conditioned* as the working formula; §6.3.10 found cumulative superego-authored prompt rewrites behaviourally inert; the green room now shows that raising authorship to judge-tier, making elicitation dialogic, and holding memory to craft-grade curation standards does not move the boundary — authorship quality and memory hygiene were not the binding constraints. The insight-action gap (§6.3.9) thereby replicates at the training level against its strongest remedy yet, and the failure localizes: not knowledge of *what* to do, but noticing *when*. The successor leverages this licenses — compiling notes into the stub’s register-policy layer (whose trigger quantities, field/DAG velocity and the stagnation composite, already exist), inserting rehearsal between note and performance, and side-coaching that delivers the note at the moment its condition holds (the referee-recomputed shape §6.15 recommends) — are recorded as future pre-registrations, not retries.

*Status and caveats (per §5.12.6).* Descriptive throughout; one profile, one world, one learner profile, one model stack. Per-note compliance pools are small (pre-note pools  $n = 1\text{--}4$ , post-note pools  $n = 0\text{--}6$  binary observations, strict n/a discipline, haiku judge): blunt enough to hide a small effect, not the  $3.4\times$  miss recorded — the pre-registered bet was that uptake

would clear 60% under exactly these conditions. All roles share one vendor lineage after the re-pin; any successor positive requires the documented cross-lineage rejudge. Exports-only (no DB writes, no rubric scoring, no human-learning claim). Artifacts committed together: the pre-registration with dated amendments (`GREEN-ROOM-PLAN.md`), gate reports (`exports/greenroom-gate0-2026-07-12/`, `exports/greenroom-gate1-2026-07-12/incl. gate1-report.{md,json}` and a sha256-manifested raw bundle), the profile with full ledger and book versions (`data/greenroom/profiles/marrick-ps-a/`), the substrate and its 26 hermetic tests (`services/greenroom/`, `scripts/greenroom*.js`, `patches/preconscious-prompt-book-context.patch`), and the diagnosis note (`notes/2026-07-12-greenroom`). Total non-subscription cost \$0.

## 6.18 First-Draft Typed Performance: a Four-Corner Boundary at Utterance Grain (Post-Hoc, Development-Tier)

The finest-grained test of the substitution law (§7.11) was not designed as one. Across roughly thirty-seven recorded iterations on 2026-07-16/17 (the “first-draft” series, V17–V53, documented as dated status notes in `notes/status/`), the tutor-stub campaign asked whether the speaking tutor’s *unaided first draft* — no mechanical repair, no model rewrite, no deterministic fallback — can satisfy a typed performance contract: the turn decomposed into owned, deterministically audited spans (uptake; a performance entry and response realizing a selected in-scene part and tactic; a concrete handoff). The question matters because the contract’s *repaired* form had already generalized: the V17 character-generalization matrix passed 4/4 cells with the repair machinery on. The series was, in effect, an attempt to move competence back from harness into model.

It closed with all four corners of a boundary measured, each on its own frozen, predeclared gate:

1. **Model-owned form fails the audit even when the semantics are right.** V51 (ordinary Codex transport, 0/1): the draft passed two independent blinded causal-fidelity reviews on all four criteria, and failed the deterministic recognizer because it wrote the gerund “blaming” where the typed pressure-target ontology admits only public-judgment nouns. Zero safety failures.
2. **Harness-repaired form passes the audit and reads worse.** The series’ blinded preference review found all six rejected originals preferred over their delivered repairs (mean  $-0.60$ ) — the repair machinery enforced form at the price of the visible exchange. The review is an exploratory single-reviewer instrument (one blinded Codex pass, not human raters — the source note’s own disclosure), so this corner is directional, not a preference study.
3. **Harness-compiled form passes and does not generalize.** With the load-bearing entry sentence compiled by the harness, the contract passed its development screen (V52, 2/2) and its home strict cell (V53, Tallow 4/4) — then rejected the first draft in all three transfer worlds, with four failure instances spanning all three owned spans: Ravensmark on both a verbatim learner-echo in uptake *and* an unrealized actorial part in performance; Skyway on an unanswered uptake slot; Foxtrot on a missing `close_inquiry` handoff action. A zero-call close-out re-audit

(notes/status/2026-07-17-first-draft-v53-closeout-diagnosis.md) reproduced all seven V53 verdicts byte-exactly and reduced these four instances to two causes: three are closed surface-form lexicon misses that only the home world’s *dictated* surfaces satisfied (a missing quotation mark; a connective outside the action lexicon; a bare-noun adjacency broken by the world’s own steering term), and one — the learner-echo — is a genuine generation defect independent of any recognizer. Either way the corner holds: extending compiled surfaces or per-world lexicons across worlds is hand-authoring the tutor per world  $\times$  span  $\times$  action.

4. **Widening the recognizer to accept semantic equivalents makes the audit vacuous** — refused prospectively by the series’ own discipline (V51: “would turn an unintentional first-draft surface into a pass and repeat the brittle audit widening the workflow is meant to avoid”).

The series verdict is its own: “V53 is terminal failed development evidence”; no held-out seed was ever consumed. What it adds to the programme is the utterance-grain instance of §7.11’s law and its sharpest statement: *semantic aptness lives in the model, checkable form lives in the harness, and the properties the adaptive-tutor construct needs simultaneously — apt, checkable, generalizing, and attributable to the model’s own conduct — did not co-occur at any tested configuration*. The wall is not refusal-shaped (zero safety failures across every recorded call; hostile stances executed on request throughout §6.17’s arms); it is the interface between formal contracts and natural-language surfaces, plus the attribution requirement itself: each move that secured checkability transferred authorship to the harness, and compiled authorship did not transfer across worlds.

*Status and caveats (per §5.12.6)*. Under §5.12.6’s taxonomy this whole section is post-hoc exploratory; we mark it “development-tier” because it sits below even that tier’s usual power — non-held-out screens,  $n=1-4$  originals per cell, no campaign-level pre-registration (gates were frozen per-iteration), one speaking-tutor family (Codex gpt-5.6-terra) on ordinary CLI transport, single-world compilation tested against three transfer worlds. The blinded reviews are causal-fidelity and single-LLM-reviewer preference instruments, not human raters or learning outcomes; no human-learning claim. Provenance: per-iteration frozen campaign YAMLS, retired seeds, and SHA-stamped notes under notes/status/2026-07-16-\* and 2026-07-17-first-draft-\*; synthesis and fold rationale in PLAN\_4\_0/2026-07-17-continue-or-fold.md; board card tutor-stub-first-draft-series. The preconscious branch folds on this result: its remaining planner gates close designed-not-run (their sensor and selector premises did not survive §6.17 and the state-benchmark closure), and the one discharged successor is the point-of-action coaching gate named at the end of §6.17, whose outcome is recorded in the addendum below.

**Addendum (2026-07-18): the point-of-action coaching gate — instrument failure on the stagnation channel; the insight–action gap persists directionally; enforcement moves action at a coverage cost.** The discharged successor ran and sealed in full: 80/80 pre-registered dialogues ( $4 \text{ arms} \times 2 \text{ trap profiles} \times 2 \text{ speaking-tutor families} \times n=5$ , world-005-marrick, POINT-OF-ACTION-COACHING-PREREGISTRATION.md, detector and rules frozen 2026-07-14, seed 20260714), zero exclusions, all traces provenance-stamped to the isolated launch branch (91b8a50e, the frozen implementation commit plus the safe-



mode-v1 claude-CLI context-isolation backport; seven pre-isolation dialogues were quarantined unanalyzed). The frozen verdict is an **instrument failure, not a coaching null**: the `stagnant_repeat` trigger under-fired against its pre-declared opportunity floor in 7 of 8 arm  $\times$  family blocks (placebo arms 8 and 4 vs floor 12) — partly because the fresh population stagnated less than the calibration replay predicted, and partly because the treatment arms suppress their own trigger’s precondition (the compiled arm recorded 2 and 1 stagnation opportunities): the trigger denominator is endogenous to the arm, a design lesson for any successor. With the density gate failed, the pre-registration licenses no mechanism verdict for either treatment arm in either family. The dense warrant channel (59–94 opportunities everywhere) supplies the descriptive picture: `side_coach` — the exact §6.16-recommended remedy, recognition delivered at the trigger moment with the action named but not enforced — lands *under* the modest +0.15 macro-compliance bar in both families (+0.146 sonnet, +0.142 sol; CIs straddle zero), extending the §6.3.9/§6.15/§6.16 insight–action gap to its point-of-action delivery form; `compiled_constraint` — enforcement through the typed-action layer — moves macro compliance strongly (+0.452 sonnet, CI [+0.089, +0.626]; +0.420 sol, CI [−0.033, +0.658]; an independent second-implementation cross-check reproduced every point estimate and verdict exactly, with the compiled arms’ CI lower bounds sensitive only to the degenerate-draw convention in the starved T1 channel — prereg §11) and rewires realized actions almost perfectly (all 180 compiled warrant-turns realize `answer_accountably`; smuggled-premise violations drop to zero), but the compliance definition’s surface `warrant_cue` remains unmet on most non-compliant turns — enforcement controls which action executes, not how it is phrased — and both compiled arms pay ~0.13 proof-DAG coverage at turn 16 against placebo, failing the would-have-applied coverage guardrail (sol also fails the safety guardrail; the zero-leak bar fails in every arm including controls). One anomaly recorded: side-coached sonnet was descriptively the safest cell in the study (hard-safety 0.90 vs placebo 0.30; 1 leak vs 10). Full tables, ledger, and the closed pre-registration: `POINT-OF-ACTION-COACHING-PREREGISTRATION.md` §11; machine-readable analysis `exports/tutor-stub-step4-claim-runs/compliance-analysis.json`; analyzer `scripts/analyze-step4-compliance.mjs`. Development-tier evidence; no DB or rubric changes; no human-learning claim. With this addendum the preconscious fold is complete: every gate on the branch is either closed with a sealed outcome or closed designed-not-run.

---

**Neighbouring modules:** Module 3 — The Tutor Did Not Become More Adaptive; Module 12 — More Layers Do Not Mean More Benefit; Module 16 — Training One Teaching Move into a Small Model.